

Xander Byrne

Astrophysics PhD Student
University of Cambridge

xbyrne.github.io
ajnb3@cam.ac.uk

Calculated solutions to modern challenges ➤

Education

University of Cambridge

Cambridge, 2019-23

- MSci Natural Sciences, 1st class with distinction
- Ranked **top of the class** in Astrophysics examinations
- Awarded University of Cambridge's prestigious *Holgate-Pollard Memorial Prize*
- Awarded *Wilson Prize* and *Skerne (1745) Scholarship* from St Catharine's College

King Edward VI Camp Hill School

Birmingham 2012-19

- A-levels (**all A***): Maths, Further Maths, Physics, Chemistry
- Twelve GCSEs, **all A*** / 9

Experience

PhD, Institute of Astronomy, University of Cambridge

Cambridge, 2023-

Investigating the composition of exoplanets and white dwarfs, using data-driven techniques

- Classification of white dwarf spectra using **dimensionality reduction** (Byrne et al., in prep.)
- Analysing **time-series data** for anomalous variability in brightness
- Submitted multiple proposals for telescope time to ESO and Keck
- Extracting elemental abundances of rocky bodies accreted onto white dwarfs

Centre for Doctoral Training in Data Intensive Science

Cambridge, 2023-24

- **Extensive training** in `pytorch` and modern machine learning and statistical techniques
- Courses in **data handling & pre-processing**, (un)supervised learning, Bayesian inference
- Experience in applying state-of-the-art analysis techniques: **MCMC**, **nested sampling**, **GPs**
- Trained **diffusion models** for image generation
- Small group teaching from **leading experts** in ML and statistics
- Seminar series from industry AI professionals; also **gave an invited talk** in this series

Masters' Project & Summer Internship

Cambridge, 2023

- **Secured over £3k of funding** to continue masters' project on Venus-like exoplanets
- Built Python wrappers for Fortran code; simulating surface-atmosphere interactions
- Invited talk at MPIA's *ExoCoffee* seminar
- Published in *Monthly Notices of the Royal Astronomical Society* (Byrne et al. 2024a)

Max Planck Institut für Astronomie – Summer Internship

Heidelberg, 2022

- **Discovery of three high-redshift quasars**, using **contrastive learning**
- Developed methodology to isolate **rare objects from a large dataset**

- Implemented cutting-edge ML techniques in **tensorflow**
- Selected methods ideal for **specific science case**, optimising architectures
- Explored a range of **unsupervised techniques** (e.g. VAEs, SOMs)
- Employed a range of **clustering algorithms** (e.g. GMMs, DBSCAN)
- Presented findings at MPIA's seminar series and two further seminars in Cambridge
- Published in *MNRAS* (**Byrne et al., 2024b**)
- Publicised in NSF NOIRLab Stories' blog series

Summer Schools & Workshops

STFC School on Data Intensive Science

Liverpool, 2024

Workshops, lectures, discussions, and panels from industry data scientists

- Workshop on **neural network emulators**
- Introduction to **Apache Spark**
- Industry session: panel discussion
- Tutorial on **publishing code**

MPIA Workshop Series

Heidelberg, 2022

- Intensive all-day workshops on **git**, **bash**, and Python

Other Projects

Supervising

Cambridge, 2023-

- Supervised 16 third-year Astrophysics students in General Relativity and Stellar Dynamics

CATAM Mathematics and Physics Projects

Computational projects and reports investigating astrophysics-related themes

- Ran simulations of accretion disks, analysing trajectories of individual particles
- Numerical integrations for testing cosmological models

International Chemistry Olympiad

Paris, 2019

- **Represented the UK** in the nation's four-person team at IChO 2019
- Assimilated a vast amount of knowledge in a **two-week timeframe**
- **Ranked 41st in the world**; awarded silver medal from the Monnaie de Paris

Cosmic radiation: HiSPARC Project & Conference; EPQ

Birmingham, 2018

- Constructed muon detectors to investigate sources of cosmic radiation
- **Wrote webscrapers** and analysis software in both **Python** and **MATLAB**
- Triangulating source directions for 190 000 cosmic ray events
- Presented findings at HiSPARC conference; **received Gold award** for best presentation
- Report including literature review & primary research

VDI Schülerforum

Frankfurt, 2016

- Group research project on drone technology
- Case study on transport of medicine to remote locations
- Presented project (partially in German) at Frankfurt University of Applied Sciences

Languages

- English (native)
- German, French (conversational)
- Mandarin, Italian (basic)